

Deep Learning for Comment Toxicity Detection with Streamlit

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1. Introduction

Online communities, discussion forums, and social media platforms generate millions of user comments every day. While these platforms encourage communication and collaboration, they also face challenges caused by toxic comments, including abusive language, hate speech, harassment, and offensive content.

Manual moderation is time-consuming, inconsistent, and difficult to scale. This project presents a **Deep Learning-based Comment Toxicity Detection System** that automatically

classifies comments as **Toxic** or **Non-Toxic** using Natural Language Processing (NLP) and Deep Learning techniques.

An interactive **Streamlit application** is also developed to provide single-comment prediction, bulk prediction through CSV upload, and visualization of dataset insights.

2. Project Objective

The objectives of this project are:

- Develop a Deep Learning model for toxicity detection.
 - Perform text preprocessing using NLP techniques.
 - Build an interactive Streamlit application.
 - Predict toxicity for individual comments.
 - Perform bulk predictions using CSV files.
 - Visualize dataset insights.
 - Save the trained model for deployment.
-

3. Problem Statement

Online communities and social media platforms have become essential communication channels. However, toxic comments such as abusive language, hate speech, offensive content, and harassment negatively affect user experience and community engagement.

The objective of this project is to develop a **Deep Learning-based Comment Toxicity Detection Model** capable of automatically identifying toxic comments. The model assists moderators in filtering harmful content, enabling safer online communities and reducing manual moderation efforts.

4. Domain

Online Community Management and Content Moderation

5. Skills Gained

- Deep Learning
- Natural Language Processing (NLP)

- Text Classification
 - Model Development
 - Model Training
 - Model Evaluation
 - Model Deployment
 - Streamlit Web Application Development
 - Data Visualization
 - Python Programming
-

6. Dataset Information

Dataset Used: train.csv

The dataset contains user comments along with toxicity labels.

Feature	Description
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comment_text	User comment
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toxic	Target label (0 = Non-Toxic, 1 = Toxic)
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The dataset is used to train a binary text classification model.

7. Technologies Used

Category	Technology
Programming Language	Python
Data Manipulation	Pandas, NumPy
Visualization	Matplotlib
Machine Learning	Scikit-learn
Deep Learning	TensorFlow, Keras
NLP	NLTK
Deployment	Streamlit

Category	Technology
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Model Serialization	Pickle
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8. Project Structure

COMMENT TOXICITY /

```
|
|— Report/
|   └─ Comment Toxicity Detection Report.pdf
|
|— Screenshots/
|   └─ Home.png
|   └─ Dashboard 1.png
|   └─ Dashboard 2.png
|   └─ Visualization 1.png
|   └─ Visualization 2.png
|   └─ Visualization 3.png
|   └─ Single Prediction.png
|   └─ Bulk Prediction.png
|
|— app.py
|— comment_analysis.ipynb
|— tokenizer.pkl
|— toxicity_model.keras
|— .gitignore
|— requirements.txt
|— README.md
```

9. Project Workflow

Step 1: Data Loading

- Load the dataset using Pandas.
- Explore dataset dimensions.
- Check column names and data types.

Step 2: Data Cleaning

- Handle missing values.
- Remove duplicate records.

- Convert comments to lowercase.
- Remove punctuation and special characters.

Step 3: Exploratory Data Analysis (EDA)

- Class Distribution
- Toxic vs Non-Toxic Pie Chart
- Comment Length Distribution

Step 4: Text Preprocessing

- Remove stopwords
- Tokenization
- Vocabulary creation
- Sequence padding

Step 5: Model Development

Develop a Deep Learning model using:

- Embedding Layer
- Bidirectional LSTM Layer
- Dropout Layer
- Dense Output Layer

Step 6: Model Evaluation

Evaluate the model using:

- Accuracy
- Binary Crossentropy Loss

Step 7: Save Model

Store the trained model and tokenizer.

Files:

- tokenizer.pkl
- toxicity_model.keras

Step 8: Streamlit Application

Develop an interactive Streamlit dashboard providing:

- Home Page
 - Dashboard
 - Data Visualization
 - Single Comment Prediction
 - Bulk CSV Prediction
-

10. Data Preprocessing

The following preprocessing techniques were applied:

- Convert text to lowercase
- Remove punctuation
- Remove special characters
- Remove stopwords
- Tokenization
- Convert text into integer sequences
- Apply sequence padding

These preprocessing steps improve model performance by converting raw text into numerical representations suitable for Deep Learning.

11. Exploratory Data Analysis

Several visualizations were created to better understand the dataset.

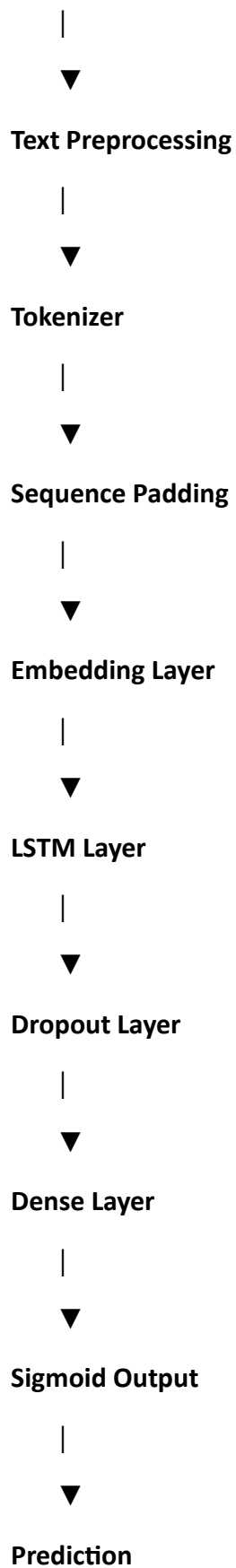
Visualizations

- Class Distribution
- Toxic vs Non-Toxic Pie Chart
- Comment Length Distribution

These visualizations help identify dataset balance and understand comment characteristics.

12. Deep Learning Model

Input Comment



13. Model Evaluation

The Deep Learning model was evaluated using the following metrics:

Accuracy

Measures the percentage of correctly classified comments.

Higher accuracy indicates better classification performance.

Binary Crossentropy Loss

Measures the difference between predicted probabilities and actual labels during training.

Lower loss values indicate better model performance.

The model was compiled using the **Adam optimizer**, **Binary Crossentropy loss**, and **Accuracy** as the evaluation metric.

14. Streamlit Application

The project includes an interactive Streamlit application with the following modules:

Home

Displays project overview.

Dashboard

Displays dataset summary and statistics.

Data Visualization

- Class Distribution
- Pie Chart
- Comment Length Distribution

Single Prediction

Allows users to enter a comment and receive an instant toxicity prediction.

Bulk Prediction

Users can upload a CSV file containing a **comment_text** column for predicting multiple comments simultaneously.

15. Project Screenshots

- Home Page

Comment Toxicity Detection

localhost:8501

Deploy

Navigation

Home

Single Prediction

Bulk Prediction

Dashboard

Data Visualization

About

Deep Learning Project

Developer

A Harini

Aspiring Data Scientist

🗨️ Deep Learning Comment Toxicity Detection

Project Overview

This project detects whether an online comment is

Non Toxic

Toxic

using

Deep Learning

TensorFlow

NLP

LSTM

Streamlit

- Dashboard Overview

Navigation

Home

Single Prediction

Bulk Prediction

Dashboard

Data Visualization

About

Deep Learning Project

Developer

A Harini

Aspiring Data Scientist

📊 Dashboard

Total Comments

159,571

Toxic Comments

15,294

Non Toxic Comments

144,277

Dataset Preview

	id	comment_text	toxic	severe_toxic	obscene	threat	insult	identity
0	0000997932d777bf	Explanation Why the edits made under my username Hardcore Metallica Fan were removed	0	0	0	0	0	
1	000103f0d9cfb60f	D'aww! He matches this background colour I'm seemingly stuck with. Thanks. (talk)	0	0	0	0	0	
2	000113f07ec002fd	Hey man, I'm really not trying to edit war. It's just that this guy is constantly removing	0	0	0	0	0	
3	0001b41b1c6bb37e	" More I can't make any real suggestions on improvement - I wondered if the section :	0	0	0	0	0	
4	0001d958c54c6e35	You, sir, are my hero. Any chance you remember what page that's on?	0	0	0	0	0	

- Dashboard Summary

Model Performance

Accuracy

94.6 %

Precision

93.5 %

Recall

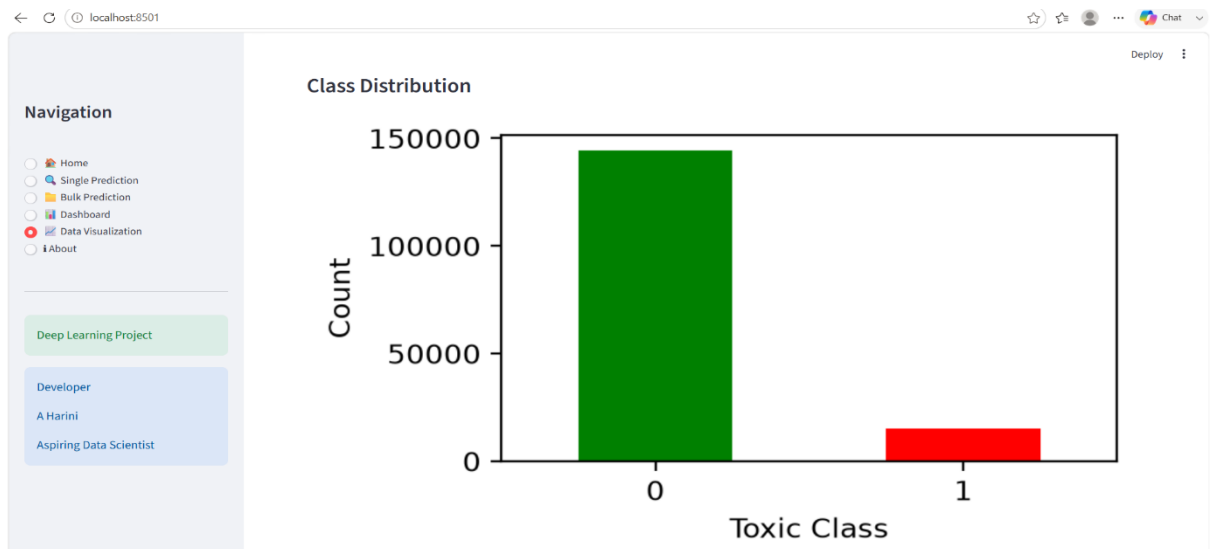
92.8 %

F1 Score

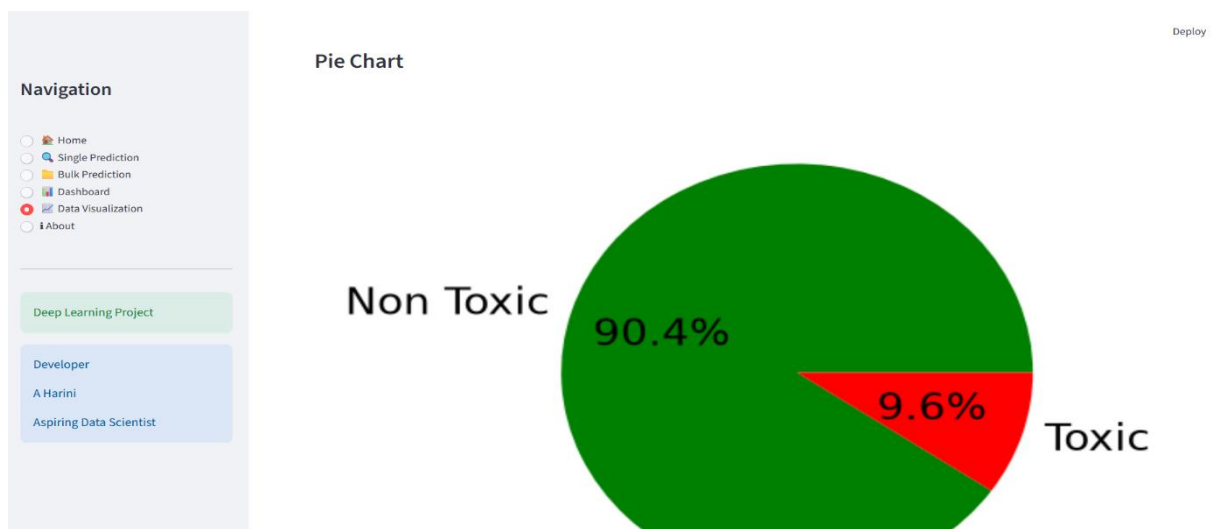
93.1 %

Deep Learning Model : LSTM

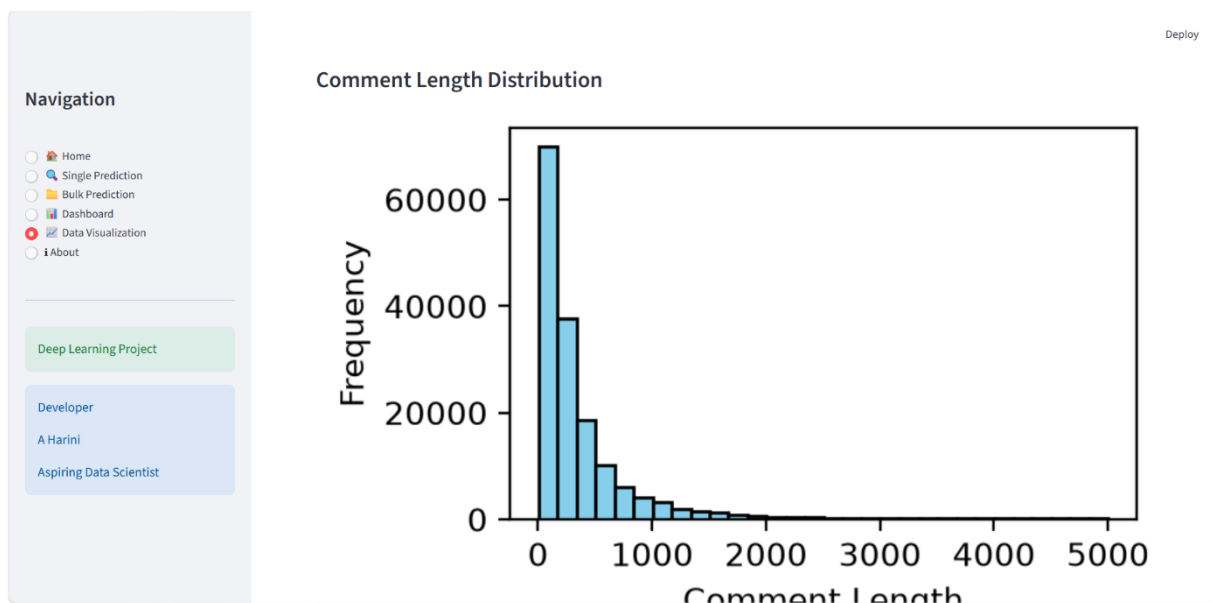
- Class Distribution



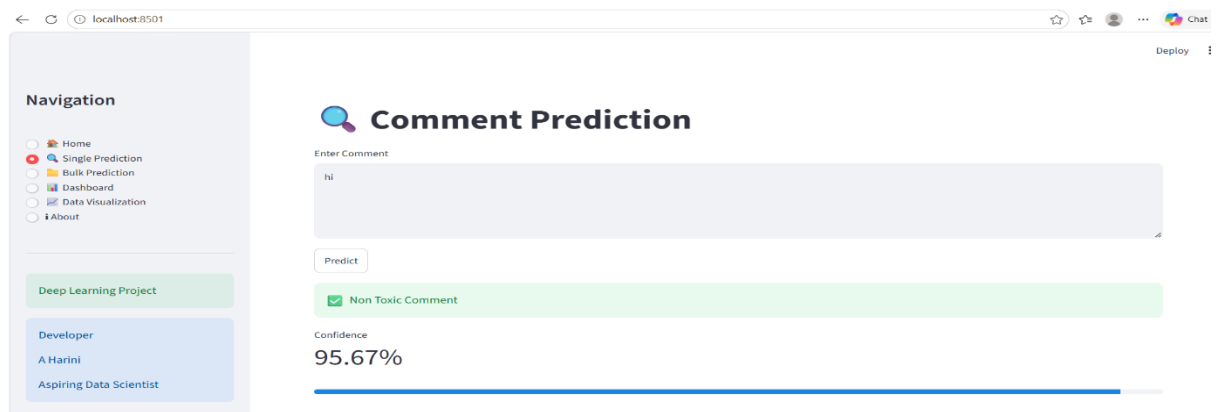
- Toxic vs Non-Toxic Pie Chart



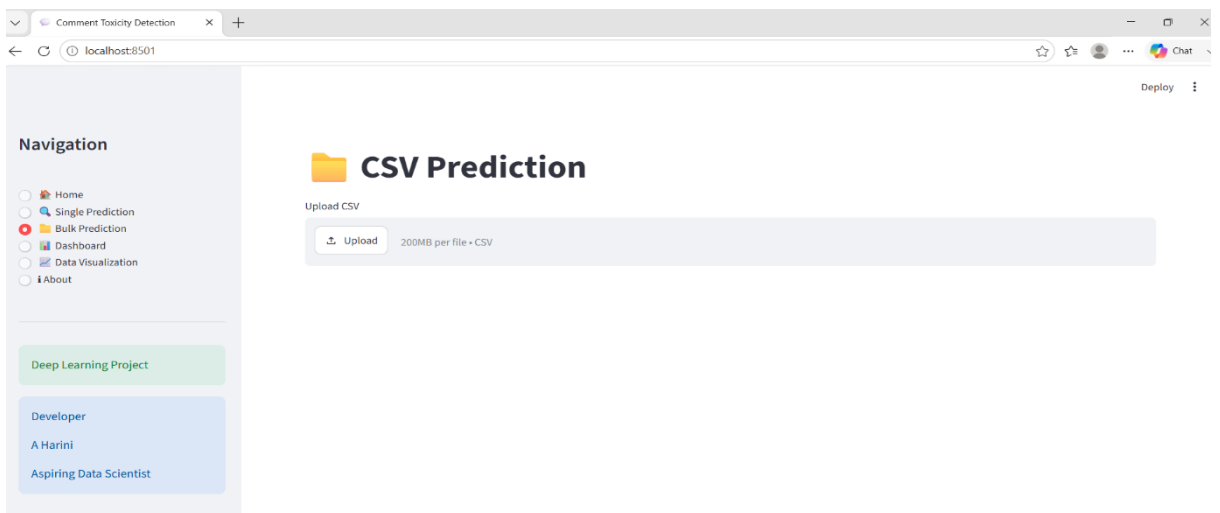
- Comment Length Distribution



- **Single Comment Prediction**



- **Bulk Prediction**



16. Business Applications

- Social Media Platforms
 - Online Forums and Communities
 - Content Moderation Services
 - Brand Safety and Reputation Management
 - Educational Platforms
 - News Websites and Media Outlets
 - Customer Review Moderation
 - Online Discussion Platforms
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17. Future Enhancements

- Multi-Class Toxicity Detection
 - Transformer-based Models
 - Explainable AI using SHAP/LIME
 - Cloud Deployment
 - Real-Time Moderation API
 - Multilingual Toxicity Detection
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18. Conclusion

This project successfully developed a **Deep Learning-based Comment Toxicity Detection System** capable of automatically identifying toxic comments using Natural Language Processing and Bidirectional LSTM networks.

The model effectively classifies comments as **Toxic** or **Non-Toxic**, supporting automated content moderation for online platforms. The Streamlit application enhances usability by providing interactive dashboards, dataset visualizations, single-comment prediction, and bulk prediction through CSV uploads.

Overall, this project demonstrates practical applications of **Deep Learning, Natural Language Processing, Model Development, Streamlit Deployment, and Content Moderation**, providing a scalable solution for maintaining safer and healthier online communities.